

SECTION B**7 (short)**

- (a) A train is traveling forwards with velocity v . A child throws a tennis ball towards the oncoming train with a horizontal speed u relative to the ground. Assuming the collision is elastic, and the train is far heavier than the ball, at what speed v_B does the ball rebound? [4]
- (b) Three balls of masses $m_1 \gg m_2 \gg m_3$ are vertically stacked at a height h , and then dropped to the ground, as shown in Fig. 5. Assuming that the balls start with small initial separations, and that all collisions occur elastically, find the height to which the mass m_3 rebounds. (You may ignore the radius of the balls.) [6]

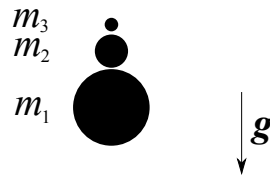


Fig. 5